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Assessment Paper AY 2023 Semester 2 E211 ESE Main (1)

Section A (12 Questions)

Q1. Figure 1a shows a representation of a queuing system. What type of service mechanism is present?

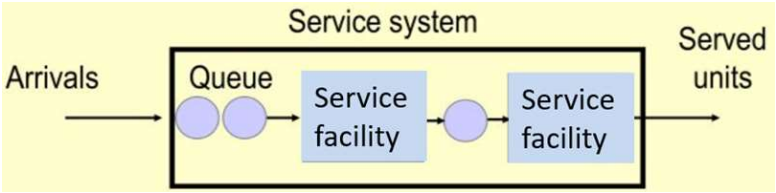


Figure 1a: Queuing System Representation

- Single channel, multi-phase (Correct Option)
- Single channel, single phase
- Multi-channel, single phase
- Multi-channel, multi-phase

Q2. In the FlexSim 3D object *Combiner* shown below, which of the following is NOT a valid Combine mode?

Figure 1b: FlexSim 3D Combiner Object

- Pack
- Batch
- Join
- Merge (Correct Option)

Q3. Referring to the partial process flow FlexSim model in Figure 1c, what would happen if 'Release cashier' activity is removed?

Figure 1c: A Partial Process Flow Model

- Payment process will take longer to complete.
- No token will flow through Payment process.
- Cashier will not be released once acquired by a token. (Correct Option)
- The simulation will not run at all.

Q4. In which category in FlexSim Healthcare library can the 3D object 'Portable Xray A' be found?

Figure 1d: Portable Xray A Healthcare Object

- Location
- Transport
- Equipment (Correct Option)
- Prop

Q5. Applying Nearest Neighbour Method to the network in Table 1, the routing solution (**starting node = B**) would be:

Table 1: Travel Distance between Nodes

	A	B	C	D	E
A		19	11	30	12
B	19		15	22	20
C	11	15		33	17
D	30	22	33		28
E	12	20	17	28	

- A – C – B – E – D – A
- B – C – A – E – D – B (Correct Option)
- B – A – C – E – D – B
- B – C – A – E – D

Q6. Referring to Figure 1e, what time series component(s) can be observed in the chart?

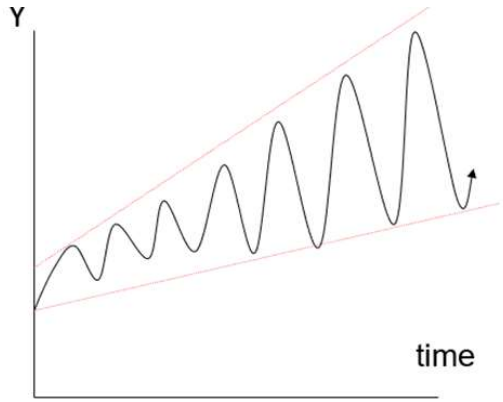


Figure 1e: A Time Series Chart

- Trend and Seasonal (Correct Option)
- Trend and Random
- Seasonal and Cyclical
- Random only

Q7. *Show your workings clearly.*

Q7a. A distribution network for medical supplies is shown in Figure 2a. The nodes represent locations and numbers on the arcs represent distances between locations.

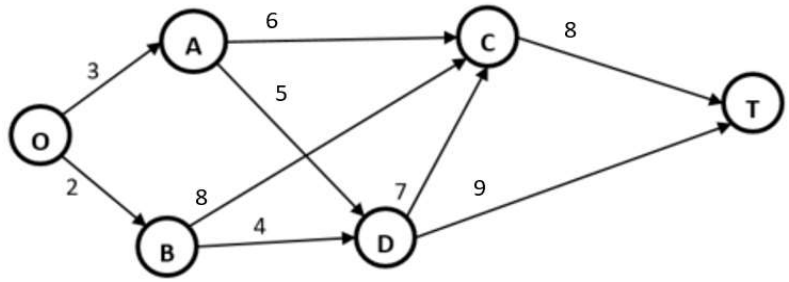


Figure 2a: A Distribution Network

Table 2a shows the partial workings to find the shortest path from node O to each of the other nodes in the network.

Table 2a: Partial Workings for Shortest Path Problem

Iteration	Solution Set	Arcs	Total Distance
Iteration 1	O	O → A O → B	3 2
Iteration 2	O B (2)	O → A B → C B → D	3 10 6
Iteration 3	O B (2) A (3)	B → C B → D A → C A → D	10 6 9 **V**
Iteration 4	…	…	…
…	…	…	…

Q7a(i). Name the algorithm used in Table 2a to solve the shortest path problem. Dijkstra's algorithm; Dijkstra algorithm;

Q7a(ii). Calculate the value of **V** in Table 2a.

Q7a(iii). Which node should be added into the solution set at **Iteration 4**? Explain.

Q7a(iv). Work out the shortest path from node O to node C and state the total distance.

Q7b. The network diagram in Figure 2b shows the distribution routes for a medical item from Plant 1 (P1) to Customer 1 (C1). The value on each arc represents the **shipping capacity** along each route.

Determine the maximum quantity that can be shipped from P1 to C1, stating clearly the route and shipment quantity along each path.

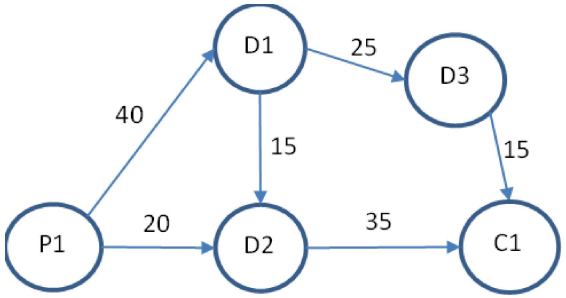


Figure 2b: A Shipping Network Diagram

Q7c. The manufacturer has two plants (A & B) producing a medical device. The product is shipped to two retailers (D & E) directly or through the transshipment hub at C. The distribution network for shipping the product is shown in Figure 2c where the numbers on the arcs represent unit shipping costs. A Linear Programming (LP) model is formulated to solve the distribution network problem in Figure 2c.

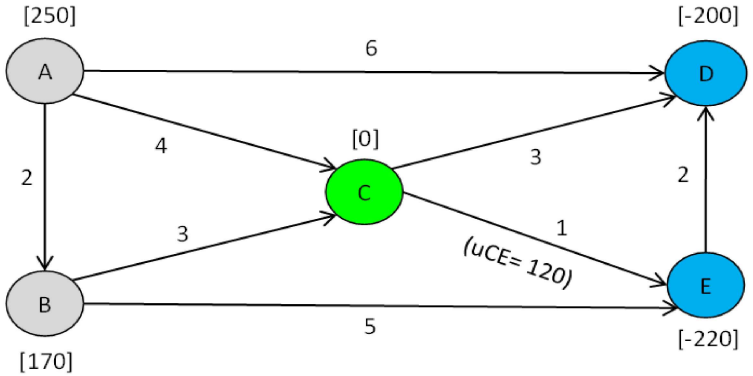


Figure 2c: Medical Device Distribution Network

Q7c(i). State in words the **objective** of formulating a distribution plan to ship the product from the plants to the customers.

Q7c(ii). Assume Xij (decision variable) represents the quantity to ship from location i to location j (i, j can be A, B, C, D or E).

Formulate the following constraints:

a. Net flow constraint for **Plant B.** (2 marks)

b. Flow capacity limit from Transshipment Hub C to Retailer E. (2 marks)

c. Net flow constraint for **Retailer D if its demand is increased to 250.** (2 marks)

Q7c(iii). List two (2) assumptions when solving the above distribution network using a Linear Programming (LP) model.

Q7d. In one country, a product is to be distributed from Depot (A) to a number of stores (B, C, D and E). A truck will leave the **Depot (A)**, visit each store once and return to the Depot after delivery. Travel distances between locations are shown in Table 2b.

Table 2b: Travel Distances Between Locations

From	To (Travel distance in km)				
	A	**B**	**C**	**D**	**E**
A	-	24	18	20	29
B	24	-	22	31	38
C	18	22	-	25	31
D	20	31	25	-	26
E	29	38	31	26	-

Clark and Wright savings method is used to plan a delivery route. Table 2c shows the **Savings** in distance between 2 stores ranked in descending order.

Table 2c: Savings Ranked in Descending Order

Rank	Savings	From	To
1	23	D	E
2	20	B	C
3	16	C	E
4	?	B	E
5	13	B	D
6	13	C	D

Q7d(i). Calculate the savings between nodes B and E.

Q7d(ii). Referring to Table 2c, at Rank 1, a sub-tour of A-D-E-A is constructed.

What step should be carried out next at Rank 2?

Q7d(iii). State the complete route based on Clark and Wrights Savings Method and calculate the total travel distance.

Q8. *Show your workings clearly. Round your answer to 2 decimal places where necessary.*

Mr Salah, who works as a sales manager, has to project the demand of a particular product for the next month. He has the following historical data on product sales (Table 3a).

Table 3a: Monthly Product Sales Figures

Month	Historical Product Sales Figure
1	540
2	475
3	610
4	580
5	525
6	575

Q8a. Using Exponential Smoothing method,

i) Work out the sales forecast for **Month 7**. Assume smoothing constant is 0.3 and forecast for Month 6 is 547.5.

ii) Calculate the absolute percent error of the forecast for Month 7 if the actual sales figure in Month 7 turns out to be 520.

Q8b. Mr Salah wants the forecast to be more responsive to the recent data. Should he use a higher or lower value of the smoothing constant for the Exponential Smoothing method?

Q8c. Suppose the sales of the product will likely see an increase in the coming months. What is the amount of lag expected if the forecasts are made using Exponential Smoothing method (assume smoothing constant = 0.3)? Recommend a more suitable forecasting method for Mr Salah.

Q8d. For another product exhibiting seasonal demands, Mr Salah is using multiplicative decomposition method to perform forecasting. Refer to the partial workings in Table 3b.

Table 3b: Partial Workings of a Forecasting Method

Year/ Quarter	Sales Figure (Y)	Time Period (X)	Moving Average	Centred Moving Average	Seasonal Ratio
2022/ 1	1400	13	960		
2022/ 2	950	14	1028.75	'A'	'B'
2022/ 3	855	15	1092.5		
2022/ 4	910	16	1125		

Q8d(i). Calculate the Centred Moving Average for **2022 Quarter 2** (labeled 'A' in Table 3b).

Q8d(ii). Calculate the Seasonal Ratio for **2022 Quarter 2** (labeled 'B' in the Table 3b).

Q8d(iii). Given that the Seasonal Index for Quarter 1 is 1.58, compute the Deseasonalized Value for **2022 Quarter 1**.

Q8d(iv). Given that the linear trend in the deseasonalized values is represented by a regression model: Y = 416.1 + 41.75X, compute the seasonalized sales forecast for **2023 Quarter 1**.

Q8e. Mr Salah would like to incorporate qualitative forecasting for product demand projection. Recommend one method and briefly explain how this method can improve his sales forecasts.

Q9. *Show your workings clearly and round off your answers to* ***two decimal places*** ***where necessary.*** *Use ^ symbol to represent the power notation.*

At a self-service ordering kiosk, an average of 15 customers arrive every hour. The time each customer takes to place their order at the kiosk is a mean of 3 minutes with a standard deviation of 1 minute.

Q9a. Using Kendall's notation, state a suitable queuing model for the above queuing system. List 2 assumptions of the queuing model.

Q9b. Calculate:

i. Mean arrival rate of customers in per minute (2 marks)

ii. Utilization of the kiosk (2 marks)

Q9c. Calculate:

i. The average number of customers waiting at the kiosk, Lq. (2 marks)

ii. The average time customers spend in the system, Ws in minutes. (2 marks)

Q9d. Suppose the number of customer arrivals has increased to 90 per hour. Will the queuing system still be in steady state? Determine the minimum number of kiosks required if the mean service time remains at 3 minutes.

Q9e. State the following characteristics (assumptions) in order that a M/M/k queuing model can be applied to study a queuing system with multiple kiosks.

i. Service time distribution Negative exponential distribution; Nega

ii. Single queue or multiple queue lines Single queue line; Single line; Single; S

Q9f. Given that the average number of customers in queue in the M/M/k model, Lq is 6.86, calculate the average waiting time of customers in the queue, Wq in minutes (assume arrival rate of 90 per hour).

Q10. Figure 5a shows a snapshot of FlexSim 3D model of a queuing system at a carwash facility. Assume that the flowitem boxes represent cars. There are 2 types of bays. **Bay1 is for 'express wash'** service whereas **Bay2 is for 'wash and groom'** service.

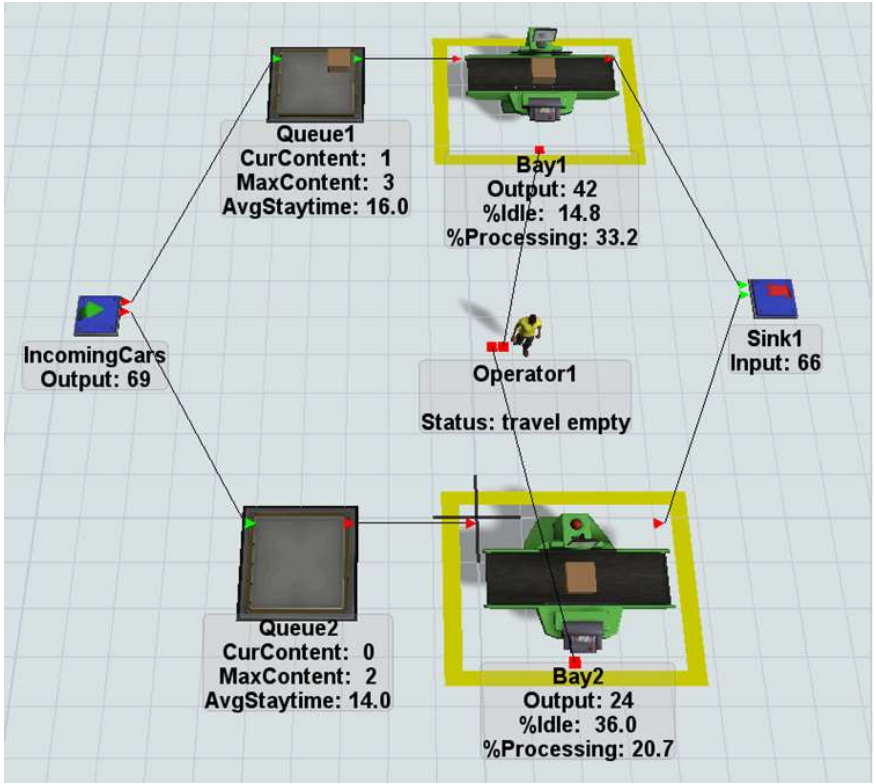


Figure 5a: FlexSim Model of Carwash Facility

Q10a. Referring to Figure 5a, state the following statistics:

i. Number of cars that have completed 'wash and groom' = 24

ii. Maximum queue length at 'express wash' = 3

Q10b. Referring to Figure 5a, suppose that Operator1 is not required for ‘express wash’, which is an automated service. Briefly describe what change(s) to make to the model.

Q10c. In ‘IncomingCars’ Source properties, under Triggers, an On Creation trigger was created as shown in Figure 5b. Describe what this trigger does.

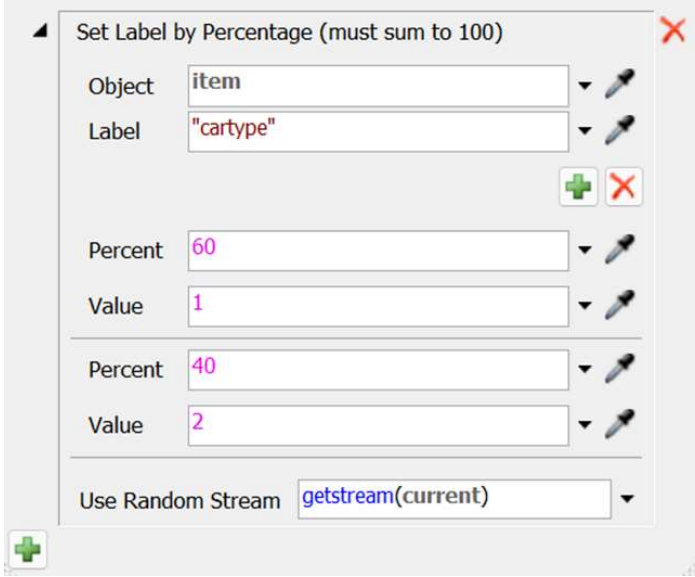


Figure 5b: On Creation Trigger Settings

Q10d. Assume that 60% of cars require 'express wash' service and 40% of cars require 'wash and groom' service. At the Send to Port option of 'IncomingCars' Source object (Figure 5c), what should be the input for each blank to simulate the flow correctly?

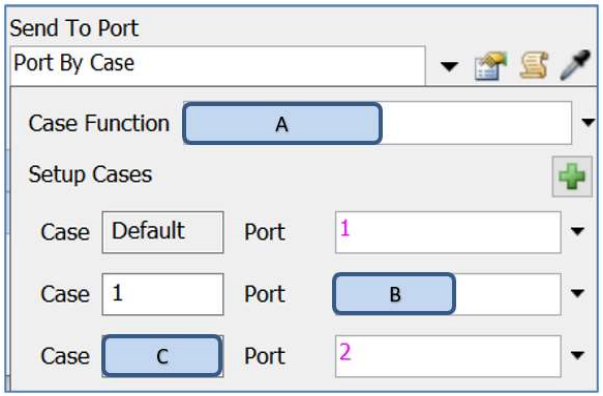


Figure 5c: Send to Port Option

A: item.cartype

B: 1

C: 2

Q11. Irene builds a FlexSim Process Flow model to study a production line comprising 2 processes – Machining and QC. Machining is carried out at Work Station while QC is conducted at QC Station.

Q11a. Figure 6a shows a snapshot of the model. Answer the following questions:

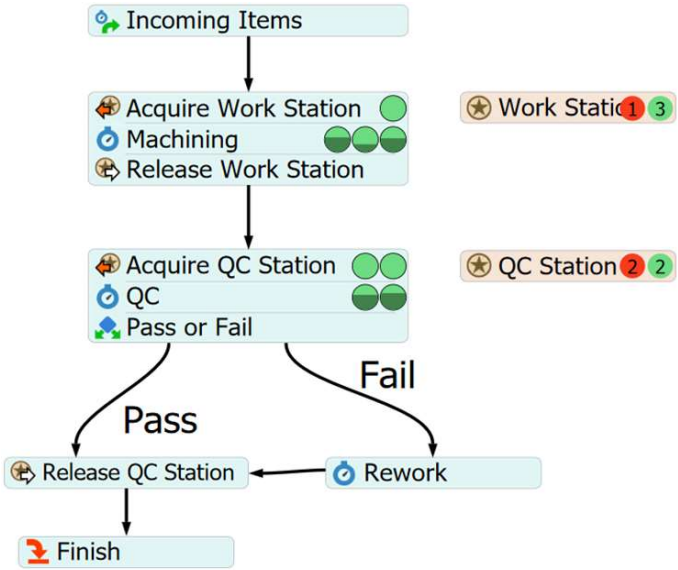
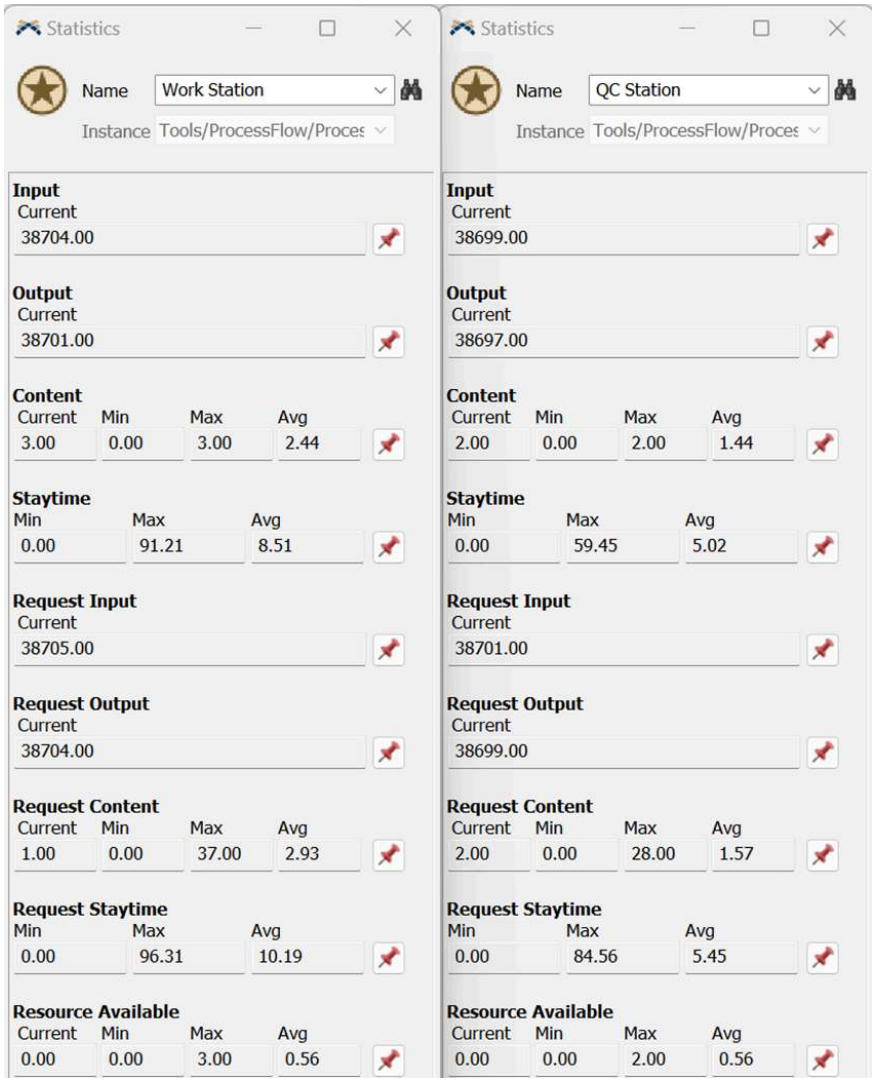


Figure 6a: A FlexSim Process Flow Model

i. State the number of items waiting for a Work Station to be available. 1

ii. Which activity (object) from FlexSim Process Flow library is used to model 'Pass or Fail'? Decide; Decide activity

Q11b. Figure 6b shows the statistics of Work Station and QC Station during the simulation. Answer the following questions:



Statistics

Name: Work Station
Instance: Tools/ProcessFlow/Proces

Input Current	38704.00			
Output Current	38701.00			
Content	Current	Min	Max	Avg
	3.00	0.00	3.00	2.44
Staytime	Min	Max	Avg	
	0.00	91.21	8.51	
Request Input Current	38705.00			
Request Output Current	38704.00			
Request Content	Current	Min	Max	Avg
	1.00	0.00	37.00	2.93
Request Staytime	Min	Max	Avg	
	0.00	96.31	10.19	
Resource Available	Current	Min	Max	Avg
	0.00	0.00	3.00	0.56

Statistics

Name: QC Station
Instance: Tools/ProcessFlow/Proces

Input Current	38699.00			
Output Current	38697.00			
Content	Current	Min	Max	Avg
	2.00	0.00	2.00	1.44
Staytime	Min	Max	Avg	
	0.00	59.45	5.02	
Request Input Current	38701.00			
Request Output Current	38699.00			
Request Content	Current	Min	Max	Avg
	2.00	0.00	28.00	1.57
Request Staytime	Min	Max	Avg	
	0.00	84.56	5.45	
Resource Available	Current	Min	Max	Avg
	0.00	0.00	2.00	0.56

Figure 6b: Statistics of Work Station and QC Station

Q11b(i). Determine the bottleneck of the process by comparing the waiting times for the 2 Stations. (staytime - waiting times)

Q11b(ii). Calculate the utilization of the QC Station.

Q11b(iii). How much space should be provisioned at the waiting area for QC stations? Assume that this should be equal to the maximum number of items waiting at QC station.

28; 28 items

Q12. Wilson, the Operations Manager of Circle Clinic created a 3D model of the clinic with FlexSim Healthcare Simulation as shown in Figure 7a.

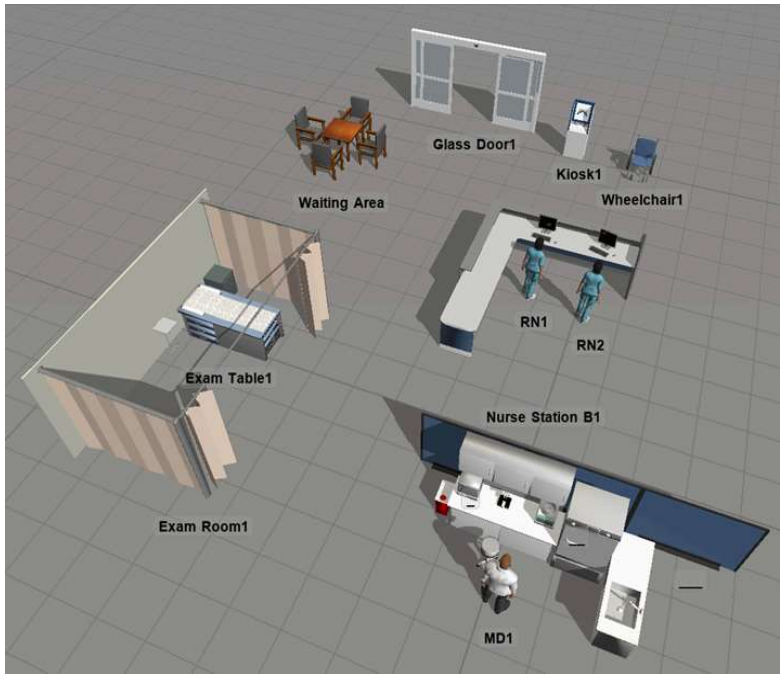


Figure 7a: Circle Clinic Healthcare Model

Q12a. Referring to Figure 7a, in which 3D object (category) of the FlexSim Healthcare library can **MD1** and **Wheelchair1** be found?

MD1 - staff

Wheelchair1 - transport

Q12b. Given the patient arrival schedule in Figure 7b, where should Wilson enter this data in the FlexSim Healthcare model?

Start (DD:)HH:MM:SS	End (DD:)HH:MM:SS	Quantity	Name
08:00:00	08:30:00	5	
08:30:00	09:30:00	3	
09:30:00	10:00:00	3	

Figure 7b: Patient Arrival Schedule

Q12c. For the flow shown in Figure 7c, recommend Wilson the appropriate HC activity sets to simulate Step 2 and Step 4.

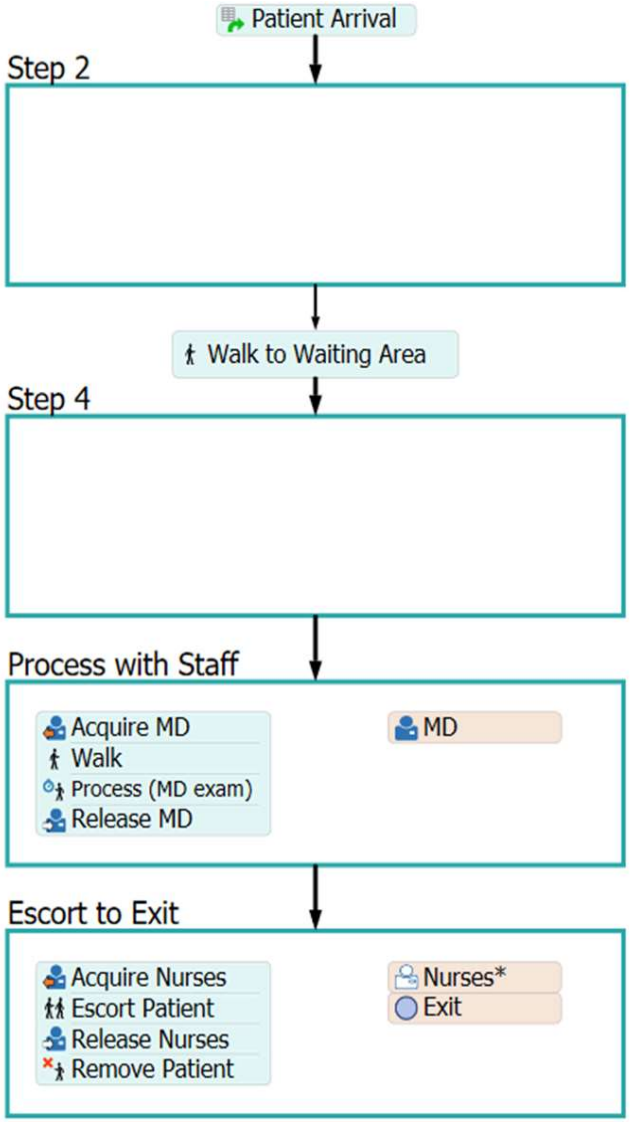


Figure 7c Incomplete Patient Flow

Step 2 - Patient walks from the glass door to the kiosk to register: Walk then process

Step 4 - Nurse escorts patient to the exam room and prepares patient for MD examination: Escort then process

Q12d. Wilson plans to create a Milestone-Milestone Times chart in a Dashboard to obtain the average time that patients stay in the clinic. Where should he insert the Milestone activities within the Patient Flow model in Figure 7c?

(Milestone can be found in the Library of Healthcare Process Flow as shown in Figure 7d)

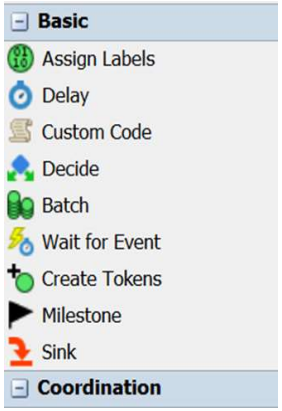


Figure 7d: FlexSim Healthcare Process Flow Basic Activities

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